

ThreatChain: Executive Summary & Explanation Guide

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1. The Problem We Are Solving

Traditional Security Operations Centers (SOCs) rely on centralized logs, which can be hacked or altered by an insider threat to hide their tracks. Additionally, when a massive DDoS attack hits, traditional blockchain security solutions fail because the network gets too congested to record the threat in real time.

2. Our Solution: ThreatChain

ThreatChain is a hybrid architecture that **decouples (separates)** the AI detection from the blockchain recording. It uses an ultra-fast XGBoost AI model to detect threats and block them at the Operating System level in milliseconds, while sending a secure cryptographic hash of the threat to the Ethereum blockchain in the background.

3. How the System Works (The 4 Layers)

- **Ingestion Layer:** Sniffs network traffic using Python (Scapy) and extracts 77 specific features defined by the CIC-IDS2018 dataset.
- **AI Inference Engine:** Uses an XGBoost model optimized with a specific mathematical loss function (logloss) to predict if the traffic is malicious.
- **Gatekeeper Logic:** If the AI is >90% confident it's a threat, it triggers two things simultaneously (Parallel Dispatch).
- **Mitigation & Ledger Layer:** It immediately runs a Windows Firewall (netsh) command to block the attacker's IP, while simultaneously emitting a low-cost LOG event to the Ethereum Smart Contract to permanently record the threat.

4. Key Results to Highlight to Examiners

- **Accuracy:** 99.24% detection accuracy with less than 0.01% False Positive Rate.
- **Speed (Sub-second mitigation):** The entire process from packet sniffing to firewall blocking takes only **~48 milliseconds**.
- **Cost Efficiency:** By using Ethereum event logs (LOG opcode) instead of state storage (SSTORE), we reduced the blockchain gas fees by over 90%.

5. How to Map the Paper to Your Code

Paper Concept	Your GitHub Code
Network Ingestion	live_sniffer.py
XGBoost AI Model	kaggle_ids2018_xgboost.py
>90% Gatekeeper Logic	evaluate_threat_confidence() function
Ethereum Connection	swarm_node.py (Web3 event listener)
Firewall Blocking	os.system('netsh advfirewall...') in Node B